

6.1 PRACTICAL EXAMPLES

MIKROELEKTRONIKA

Picture 6.5 detects when light is present in a room. With no light, the photo-transistor does not conduct. When light is present, the photo-transistor conducts and the bell is activated. Turning off the light will not stop the alarm. The alarm is turned off via S1.

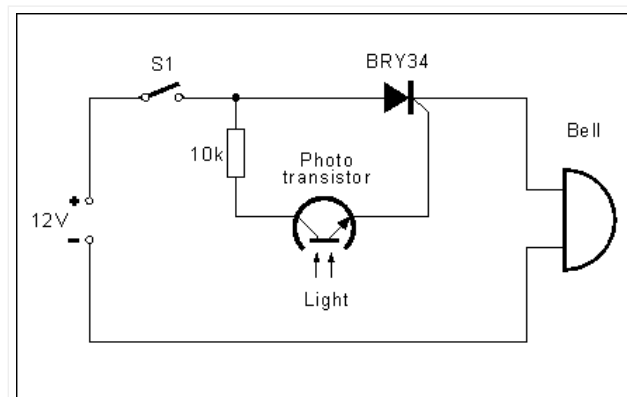


Fig. 6.5: Alarm device using a thyristor and a photo-transistor

A circuit to flash a globe is shown in figure 6.6. This circuit flashes a 40w globe several times per second. Mains voltage is regulated using the 1N4004 diode. The 220u capacitor charges and its voltage rises. When this voltage reaches the design-voltage of the diac (20v), the capacitor discharges through the diac and into the triac. This switches the triac on and lights the bulb for a very short period of time, after a period of time (set by the 100k pot), the capacitor is charged again, and the whole cycle repeats. The 1k trim pot sets the current level which is needed to trigger the triac.

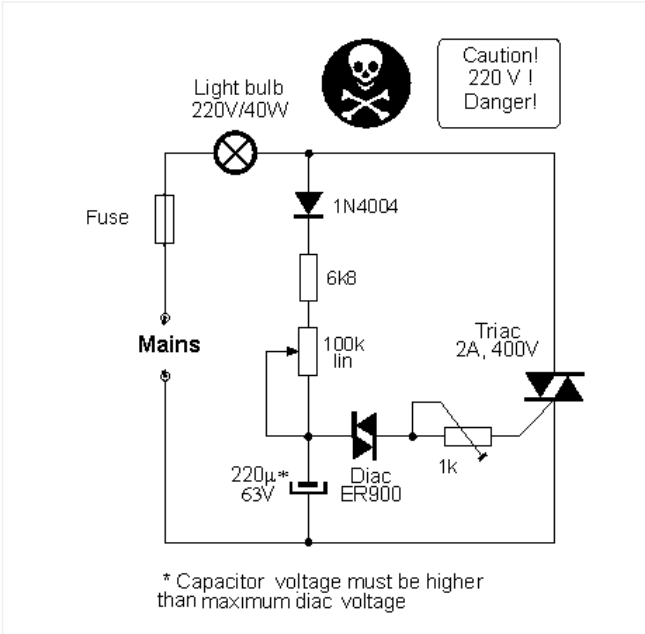


Fig. 6.6: Flasher

A circuit to control the brightness of a globe or the speed of a motor is shown in figure 6.7

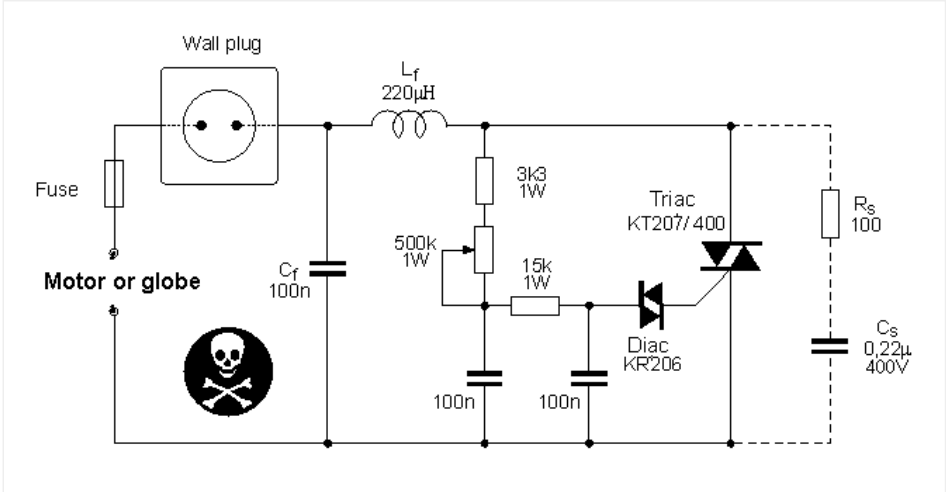


Fig. 6.7: Light bulb intensity or motor speed controller

If the main use for this circuit is to control the brightness of a light bulb, RS and CS are not necessary.



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